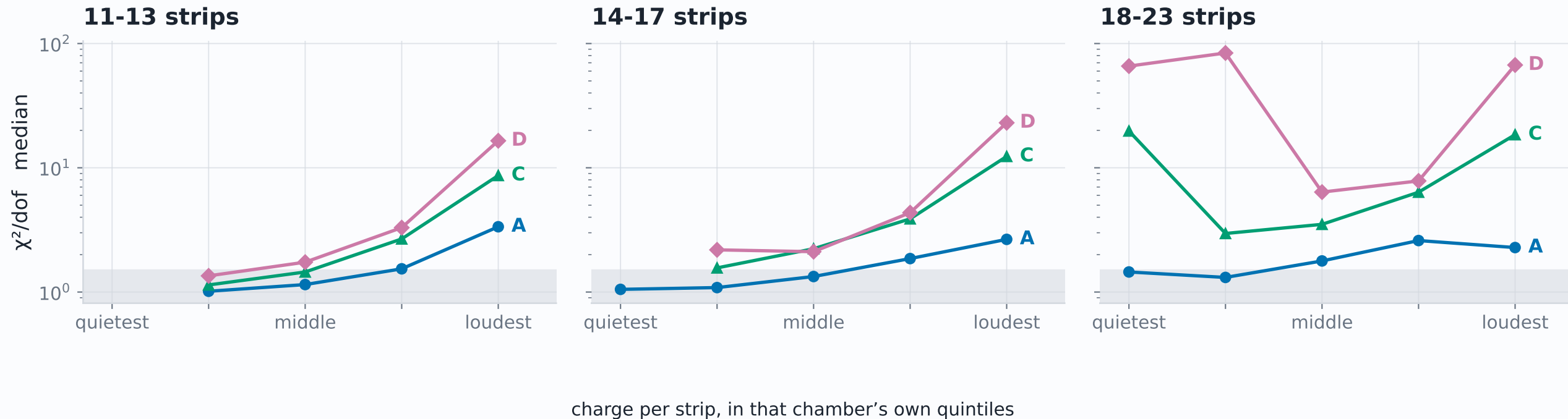


At fixed track length, the loud tracks are the bad fits



Quintiles are per chamber, because the gains are not cross-calibrated — a bin means "this chamber's quietest fifth", never a shared charge. This is the mechanism behind the length trend: the model's residual is a fixed FRACTION of the pulse, so on a quiet track it is buried in the noise and χ^2/dof sits on the floor, while on a loud one it outgrows the noise. A's low- χ^2 tracks carry $\sim 2.5\times$ less charge than its high- χ^2 ones at identical n_{strips} . The upturn at the quiet end of C and D in the third panel is the other failure — a fit that explained nothing — which this selection mostly, but not entirely, removes.